

Analysis of bias

A quick visual reference for spotting systematic error in a study

Bias

A systematic, not random, deviation of results from the truth, built into how a study was run.

Internal validity

Whether the study's own results can be trusted, before asking if they generalize elsewhere.

Four major bias families

Selection bias

Who gets into the study isn't representative

Information bias

How data is measured or recalled is flawed

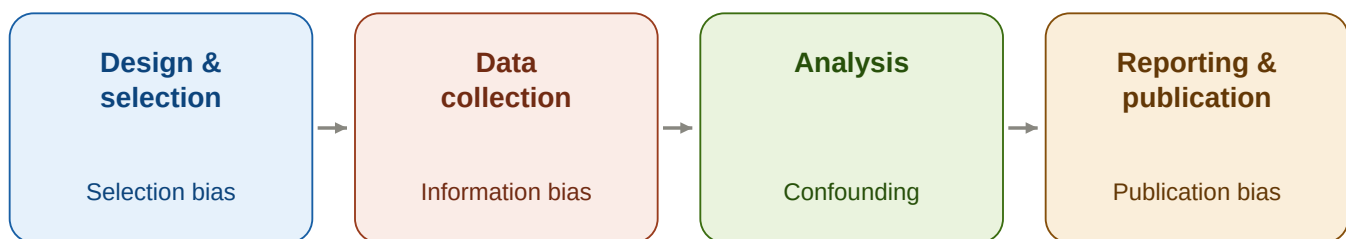
Confounding

A third variable distorts the true effect

Publication bias

Null results are less likely to be published

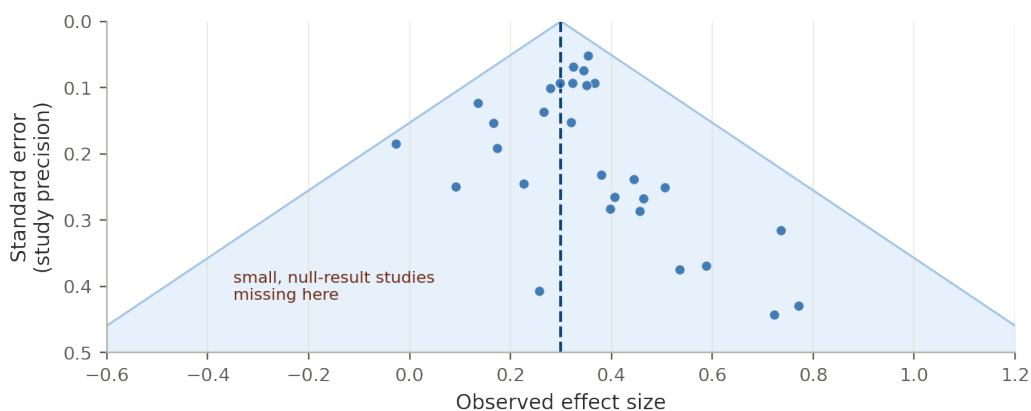
Where bias enters, stage by stage



Always ask: which way does this bias push the result?

Toward the null (hides a real effect) vs. away from the null (exaggerates or invents an effect)

Funnel plot: a classic check for publication bias



Quick reference: a symmetric funnel suggests no publication bias; a gap on one side suggests missing studies.